HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and implemented **Retrieval-Augmented Generation (RAG) pipelines** using LangChain and vector databases to enhance LLM response accuracy for customer-facing applications, processing approximately **4,500 queries** per day.
- Developed and deployed **AI agents** and workflow automation solutions with **Python**, integrating them into existing enterprise platforms to automate complex business processes, reducing manual effort by an estimated **18%**.
- Built scalable **APIs and microservices** with **FastAPI** to serve AI model inference and integrate intelligent capabilities into backend systems, achieving a **p95 latency** of under **200ms**.
- Collaborated with software engineering and data teams to translate business requirements into technical solutions for **Generative AI applications**, participating in daily stand-ups and sprint planning.
- Contributed to technical architecture discussions and conducted code reviews for **AI-powered applications**, ensuring adherence to software engineering principles and best practices.
- Researched and evaluated emerging AI technologies and frameworks, including advancements in **LLM fine-tuning** and prompt engineering, to identify opportunities for innovation.
- Deployed and maintained **AI applications** in a **cloud environment** (AWS), ensuring scalability, security, and implementing monitoring solutions to track performance and cost efficiency.

SKILLS

AI/ML – Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), AI Agents, Prompt Engineering, Model Evaluation, MLOps, LLMOps, Vector Databases, Embeddings, Fine-tuning, TensorFlow, PyTorch, scikit-learn

Programming Languages – Python, Java, SQL, R

Frameworks/Libraries – LangChain, LlamaIndex, Hugging Face, FastAPI, Flask, Spring Boot

Cloud Platforms – AWS (EC2, S3, Lambda, CloudWatch), Azure OpenAI, Amazon Bedrock, Google Vertex AI

Development Tools – Docker, Kubernetes, Git, Jenkins, Jira, Confluence

Data Management – PostgreSQL, MySQL, Oracle, Data Warehousing, ETL, Data Modelling

Observability & Monitoring – Splunk, Dynatrace, CloudWatch

PROJECTS

Customer-Facing AI Assistant with RAG

- Developed a **Generative AI assistant** for enterprise customers, implementing a **RAG pipeline** that integrated with internal knowledge bases via vector embeddings.
- Built a scalable **Python backend** using FastAPI to serve the AI assistant's API, handling approximately **3,000 requests** daily and processing unstructured data from customer support tickets.
- Integrated the AI assistant into an existing customer portal, providing a conversational interface for technical inquiries and reducing average response times by **22%**.

Intelligent Workflow Automation Service

- Engineered a **Python-based AI service** to automate reconciliation workflows for financial transactions, significantly reducing manual data validation efforts.
- Designed and implemented **LLM-powered agents** to identify and flag discrepancies within transaction logs, achieving an accuracy rate of **91%** on a dataset of over **100,000 records**.
- Containerized the service using **Docker** and deployed it on **AWS Lambda**, ensuring high availability and efficient resource utilization for processing high volumes of transactions.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)